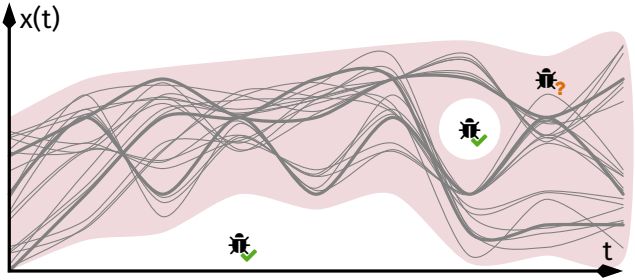

# Abstract Interpretation

Software Quality Assurance — Static Code Analysis, II | Florian Sihler | December 3, 2025

Software Engineering
Programming Languages

universität ulm

# 1. The Why

Why am i even here?

# Embedding a Landscape

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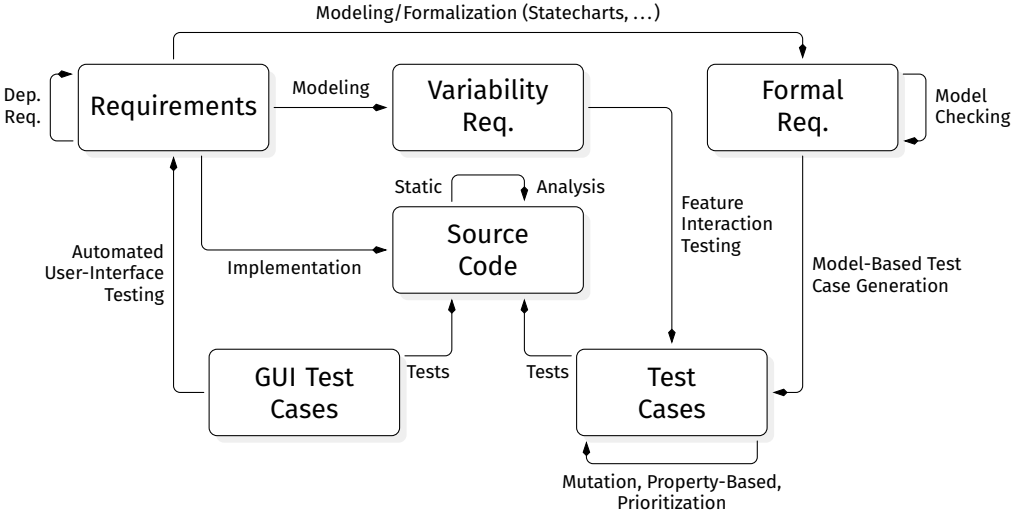

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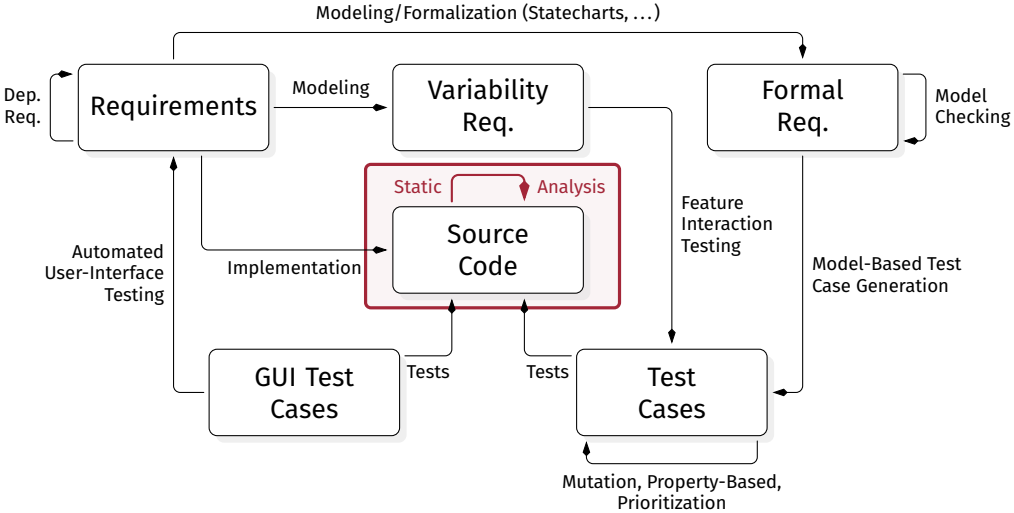

## **What** is static analysis?

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Today, we learn how to…

- describe semantic properties
- compare, refine, and combine properties
- describe and map language semantics to properties
- deal with the cost of abstraction (and the fun)

# The Why

```java
public static void main(String[] args) {
   int a = 1;
   double r = Math.random() * 10;
   if (r > 5) {
      a = 2;
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   System.out.println(1 / a);
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- We want to proof, that a program satisfies certain properties
- Abstract Interpretation is one (/the) technique to do so

# Origins

Static Analysis

Based on the amazing “Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation” by Miné [Min17], https://www.di.ens.fr/~cousot/AI/, and [Bal+18; GR22]

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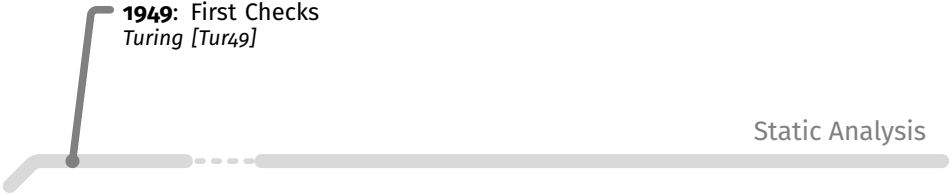


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*Non-trivial properties are undecidable [Ric53]*

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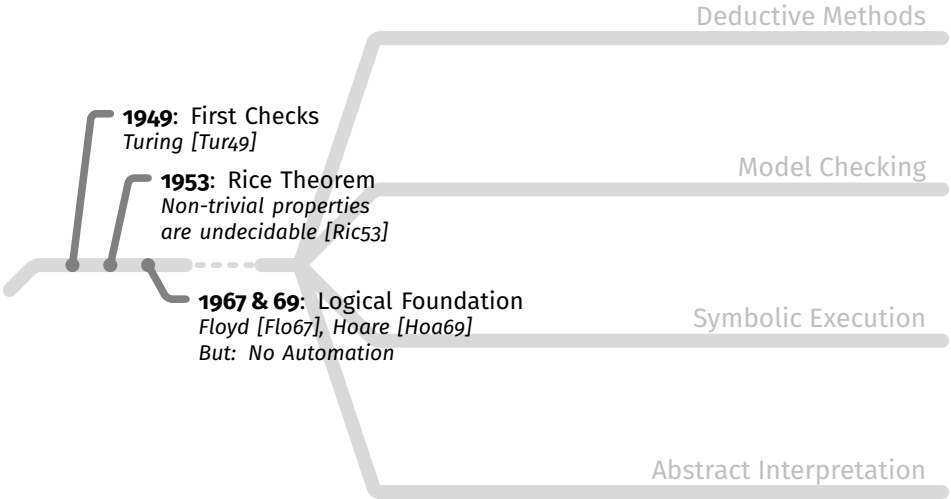


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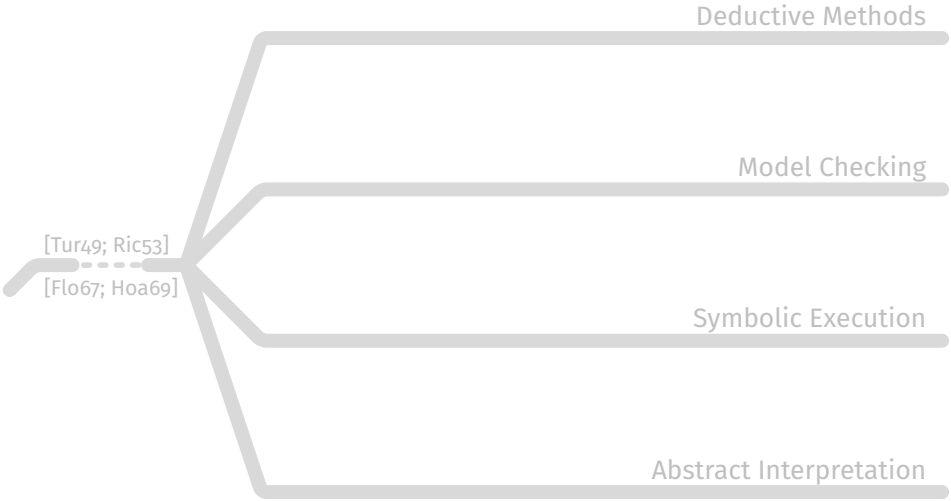


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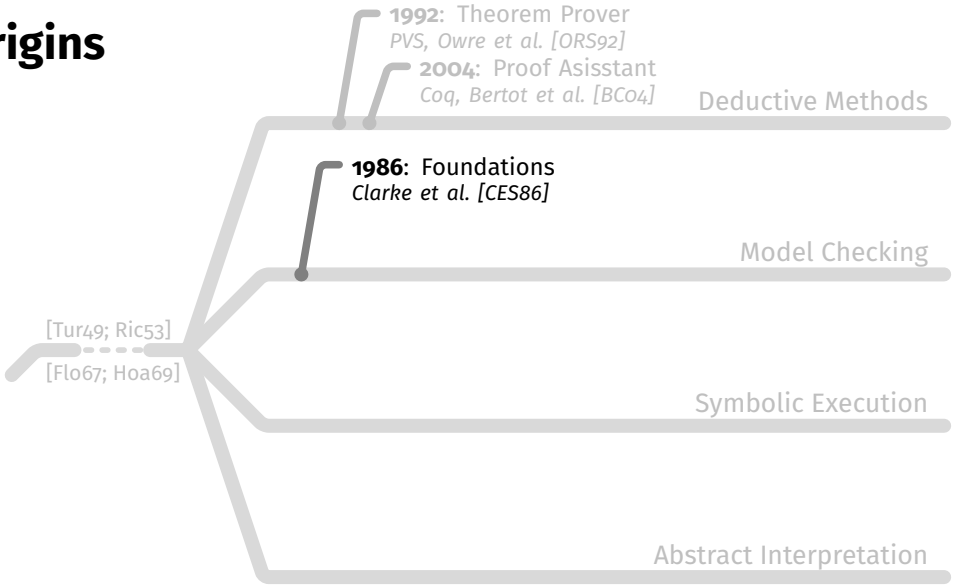


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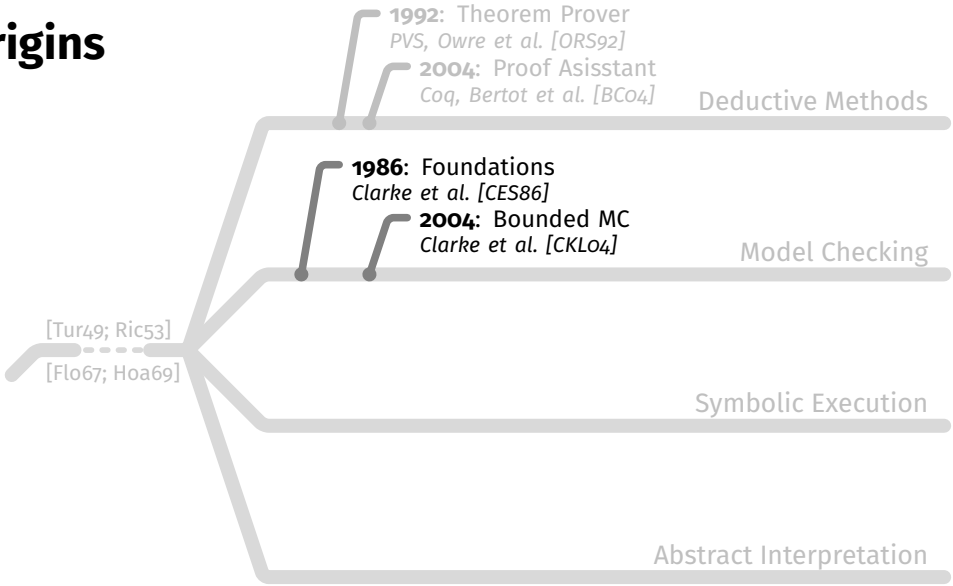


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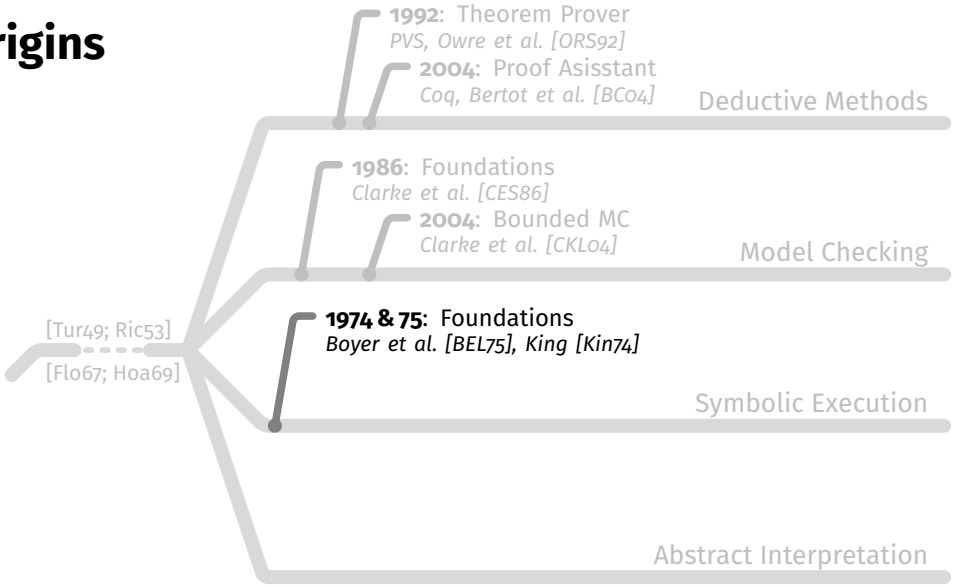


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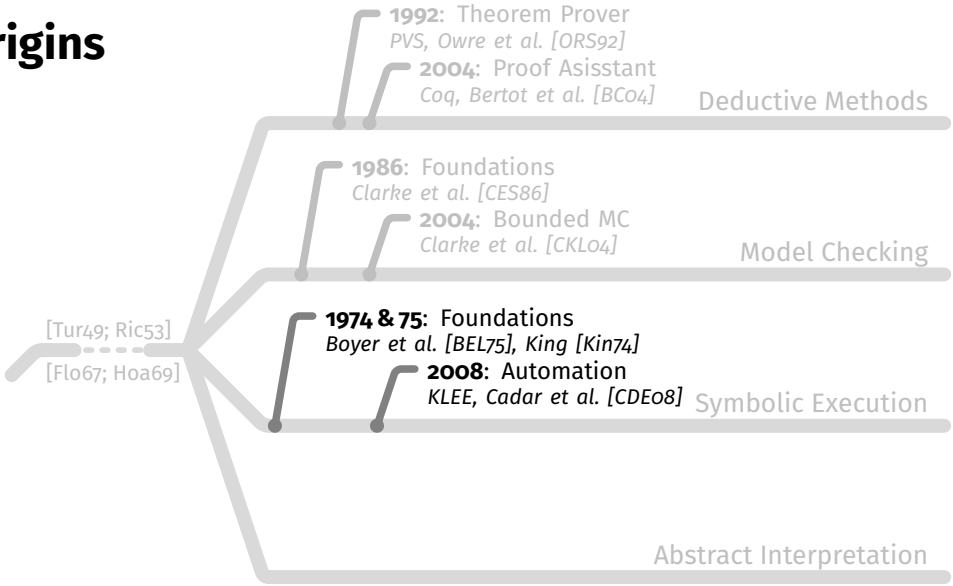


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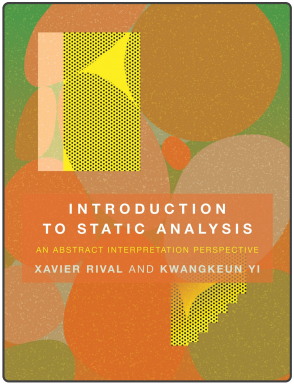


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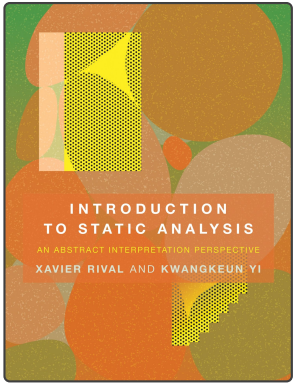


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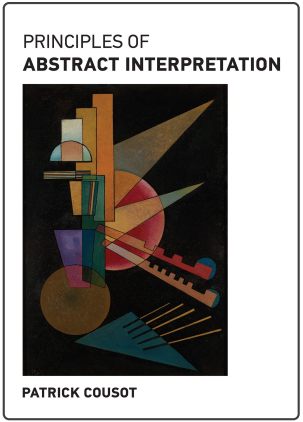


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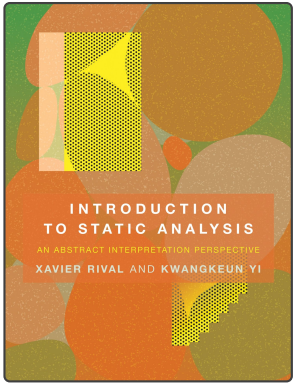


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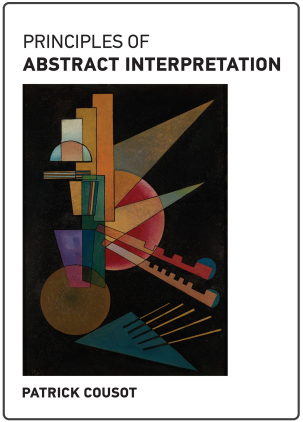


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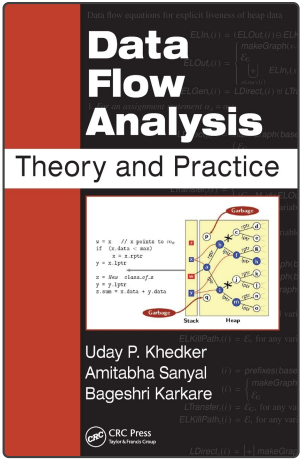


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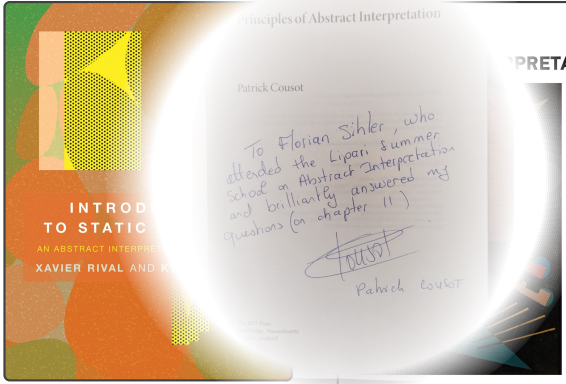


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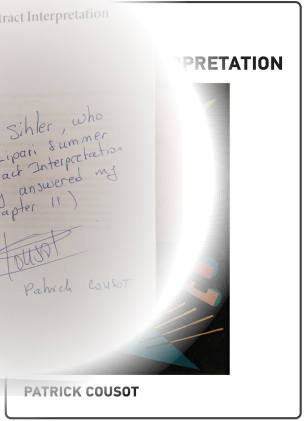


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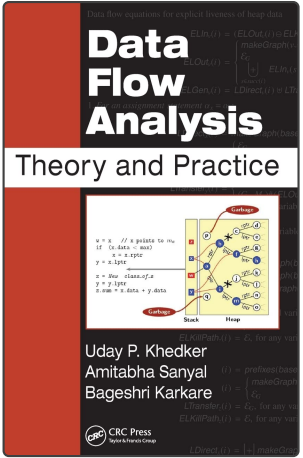


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# 2. The How

Gimme properties, gimme abstractions!

# Abstract Interpretation

See “A casual introduction to Abstract Interpretation” [Cou12]

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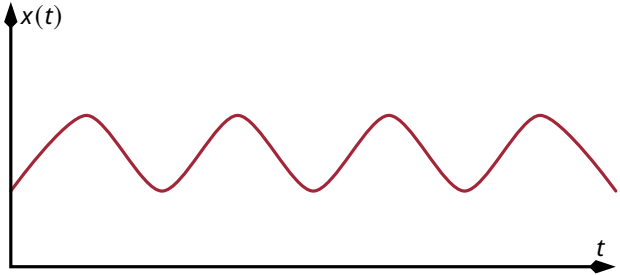


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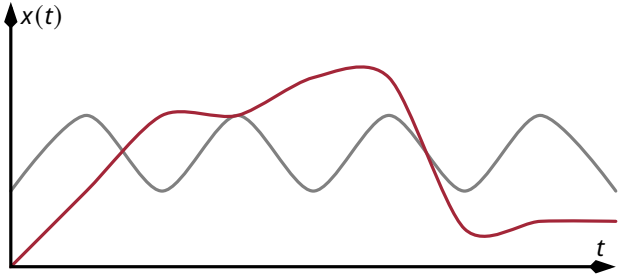


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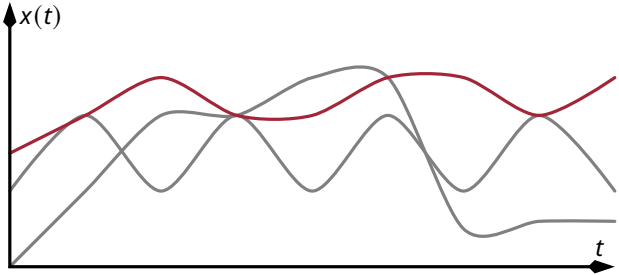


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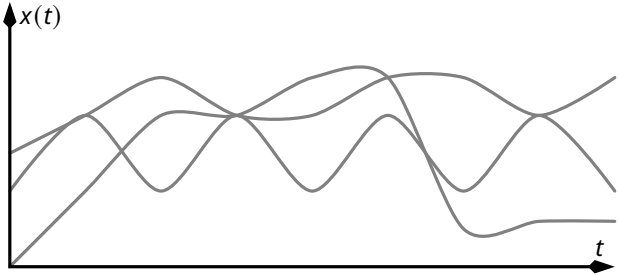


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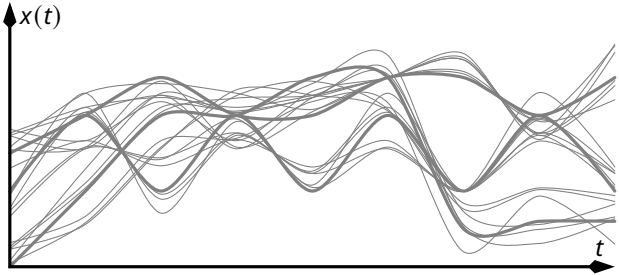


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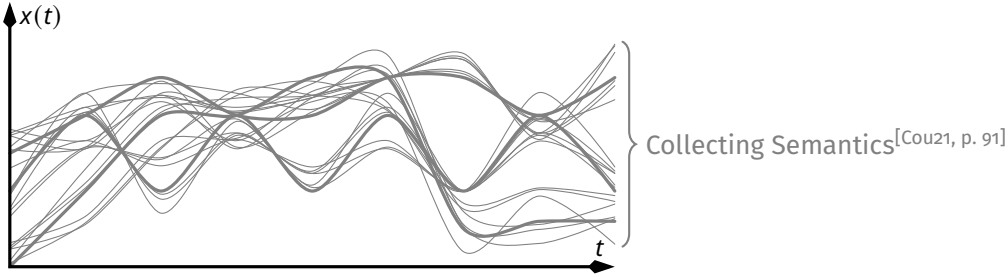


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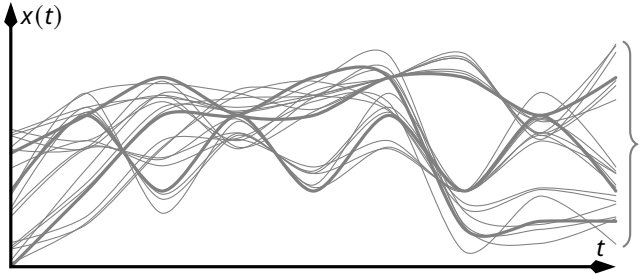


Collecting Semantics[Cou21, p. 91]

- Maybe impossible to compute statically
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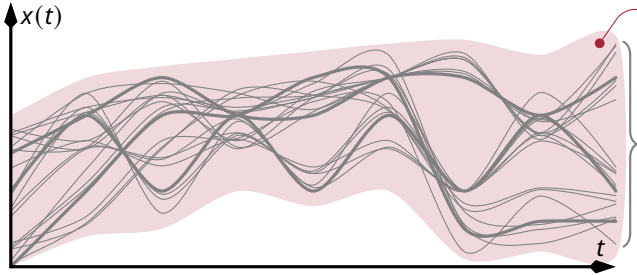


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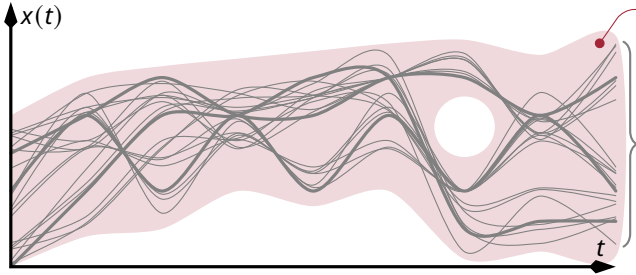


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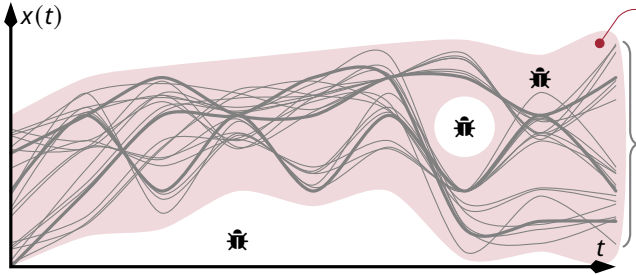


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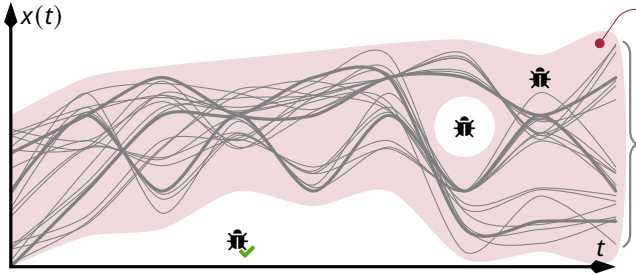


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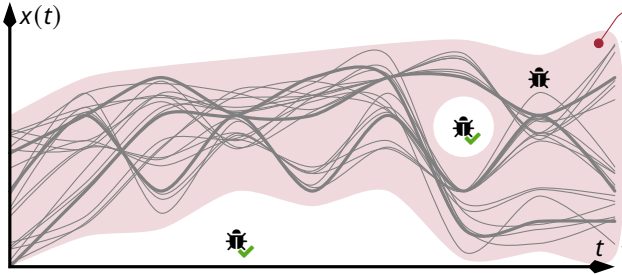


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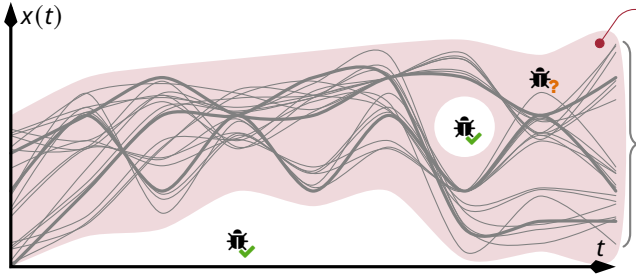


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**Radhia Cousot (1947–2014)**
Patrick Cousot

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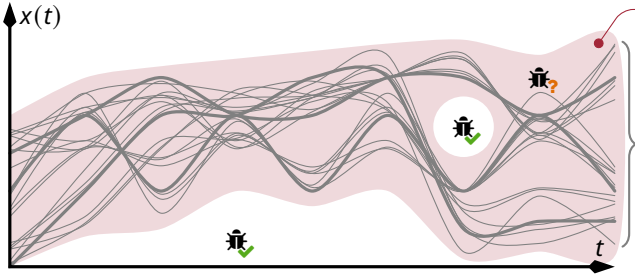


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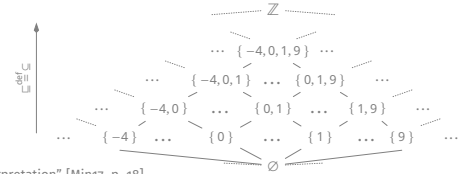
Hasse Diagram of $(\mathcal{P}(\mathbb{Z}), \subseteq)$

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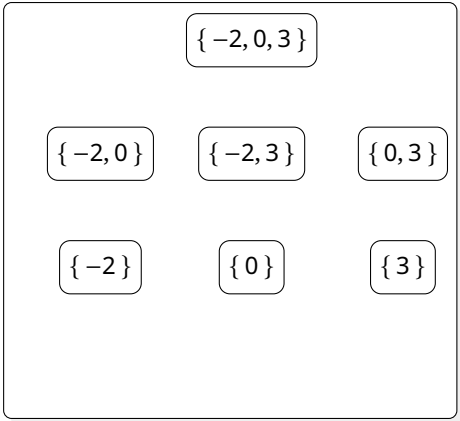


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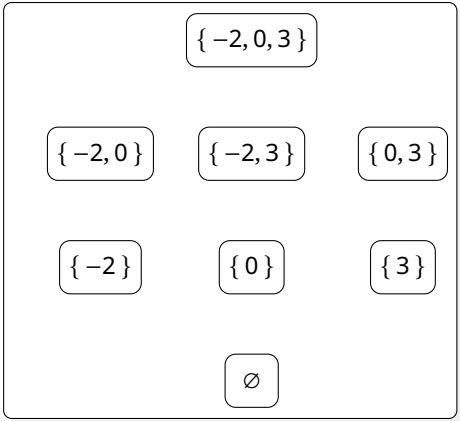


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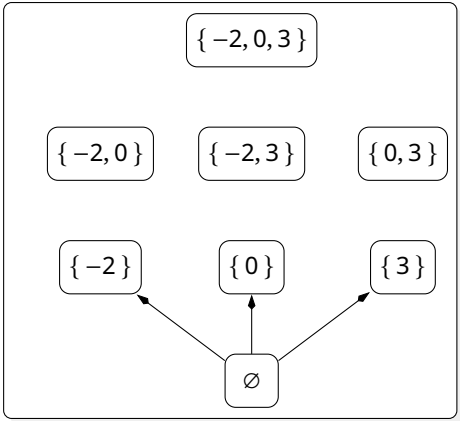


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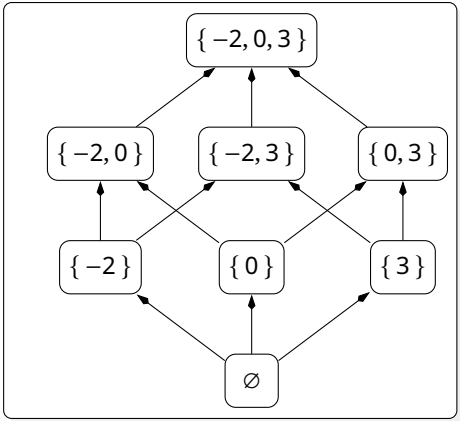


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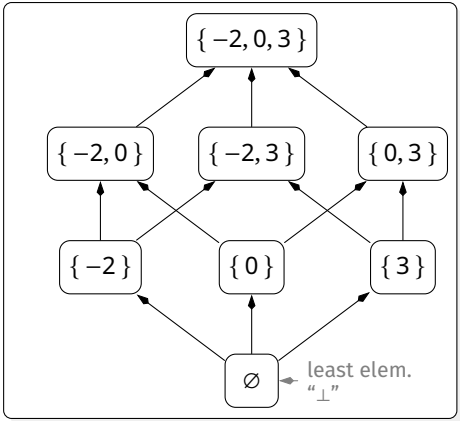


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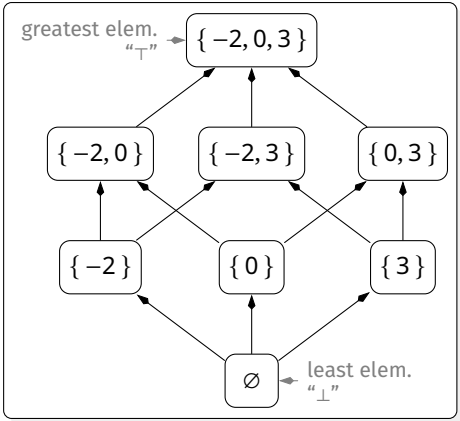


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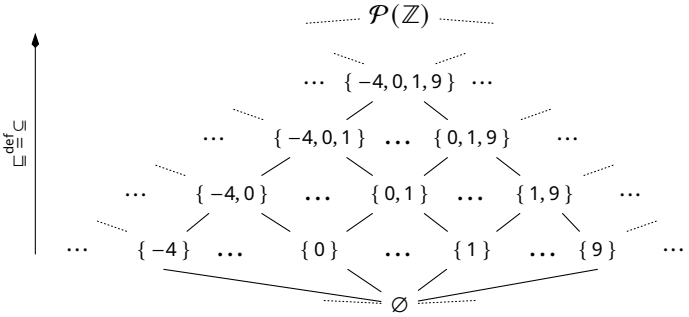


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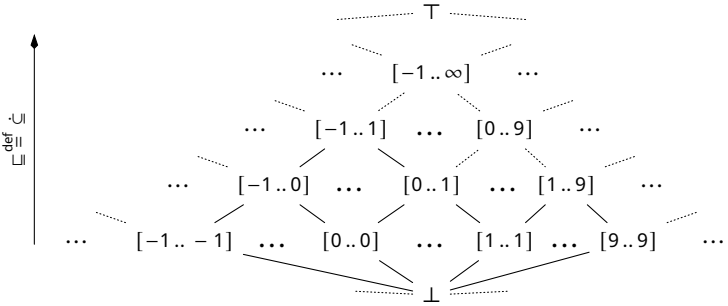


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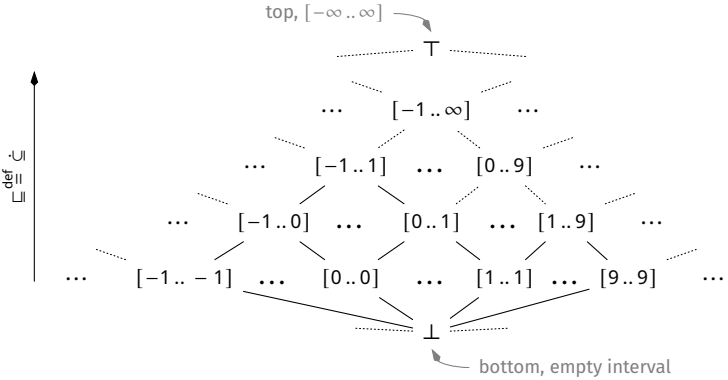


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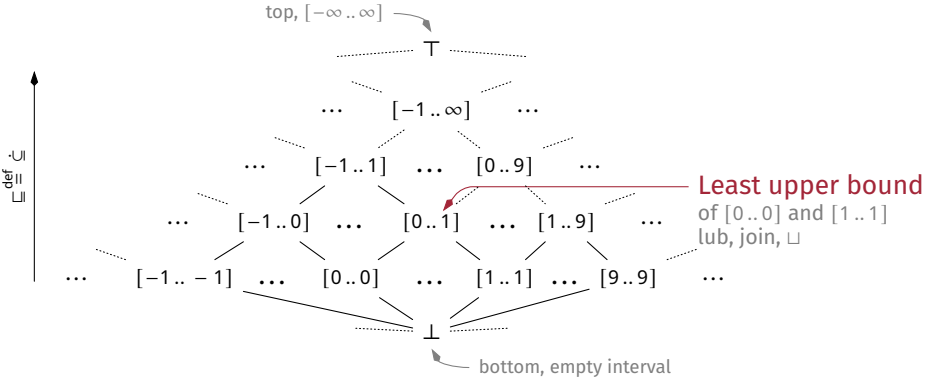


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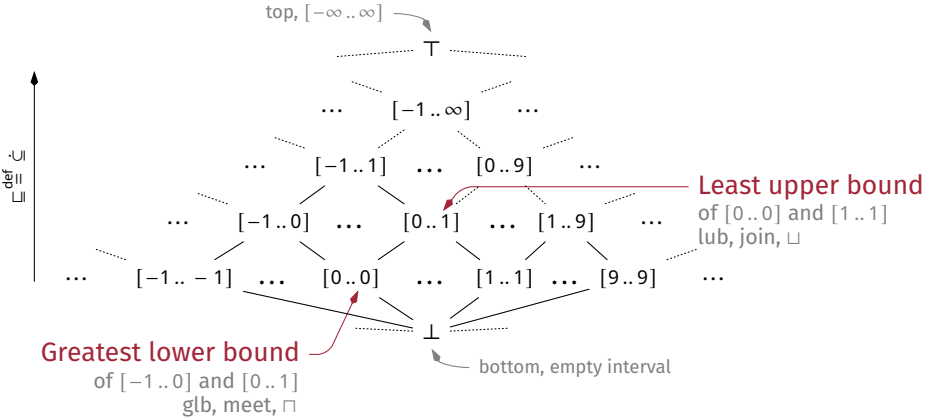


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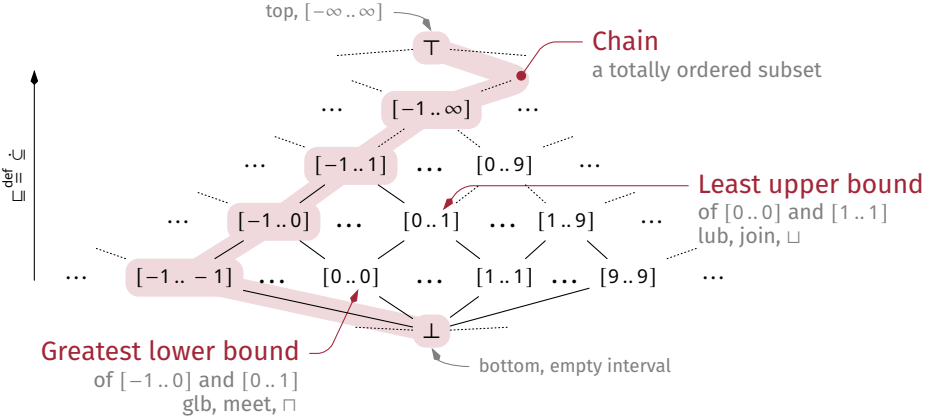


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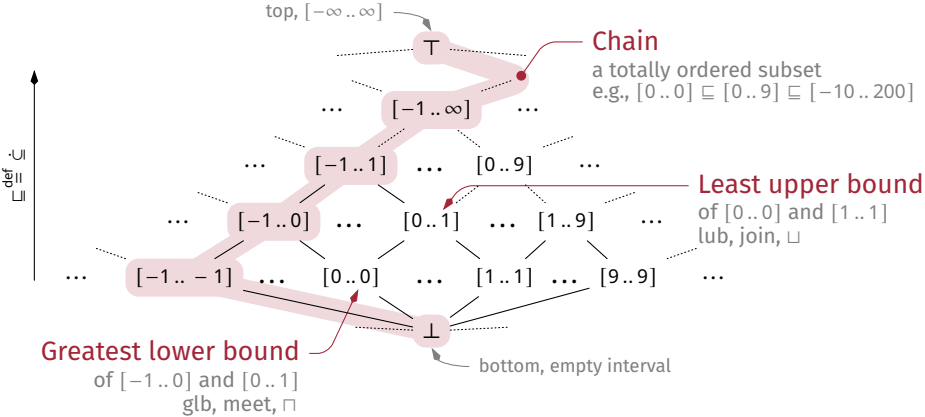


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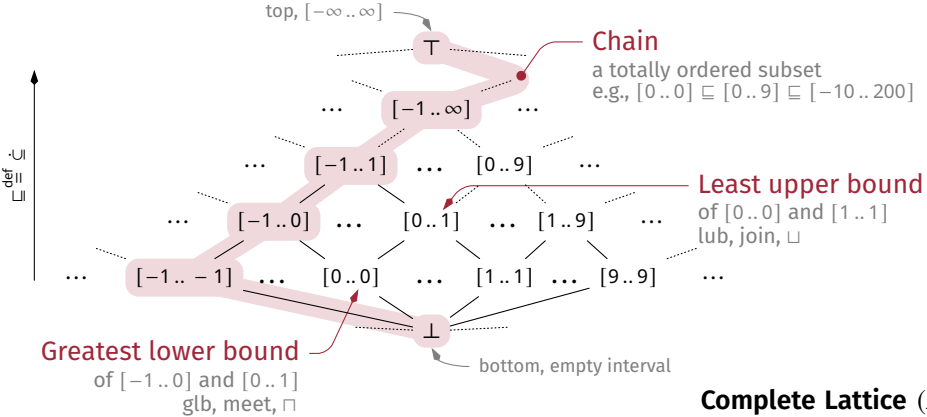


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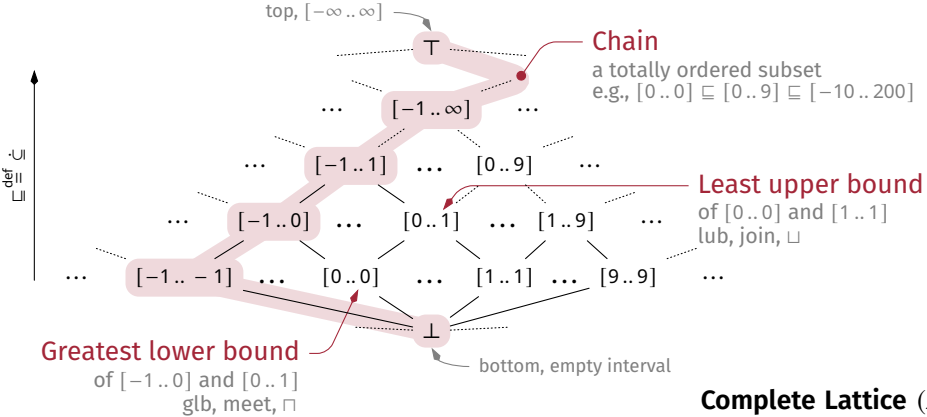


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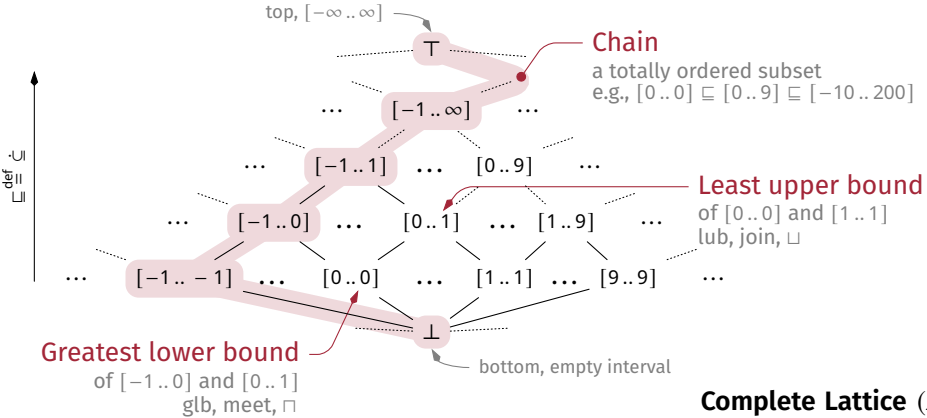
- $(X, \sqsubseteq)$ is a partial order

"Lattice theory" [Bir67], see also sublattices [Min17, p. 25]

# Posets, Lattices, and Chains

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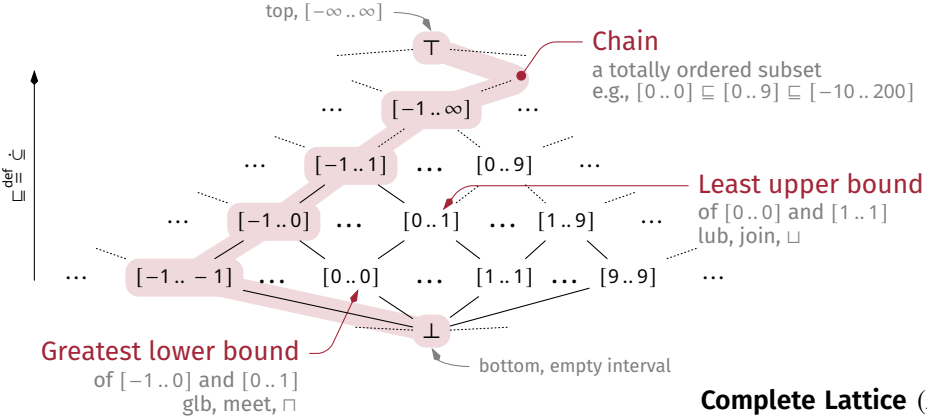
# Posets, Lattices, and Chains

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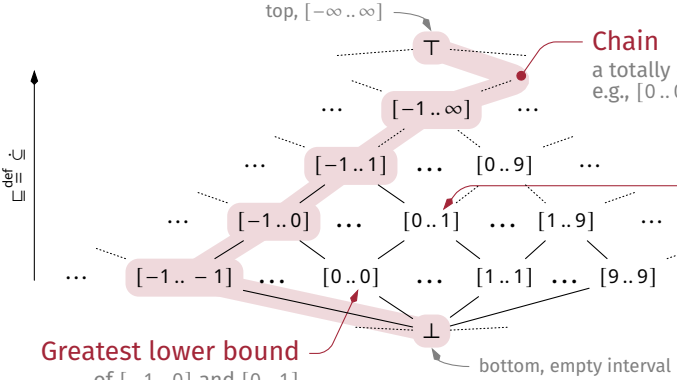
- $\bot / \top$ as smallest/largest element

"Lattice theory" [Bir67], see also sublattices [Min17, p. 25]

# Posets, Lattices, and Chains

**Garrett Birkhoff (1911–96)**
Paul Halmos




**Chain**
a totally ordered subset
e.g., $[0\,..\,0] \sqsubseteq [0\,..\,9] \sqsubseteq [-10\,..\,200]$

**Least upper bound**
of $[0\,..\,0]$ and $[1\,..\,1]$
lub, join, $\sqcup$

**Greatest lower bound**
of $[-1\,..\,0]$ and $[0\,..\,1]$
glb, meet, $\sqcap$

**Complete Lattice** $(X, \sqsubseteq, \sqcup, \sqcap, \bot, \top)$

- $(X, \sqsubseteq)$ is a partial order
- $\forall A \subseteq X : \sqcup A$ and $\sqcap A$ exist
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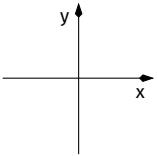
"Lattice theory" [Bir67], see also sublattices [Min17, p. 25]

# Abstract Domains

Pick Your Favorite Lattice!

## Numerical




"Principles of Abstract Interpretation" [Cou21], "Pentagons: a weakly relational abstract domain for the efficient validation of array accesses" [LF08, p. 25]

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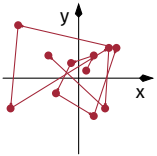


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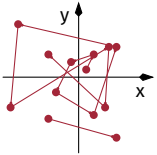


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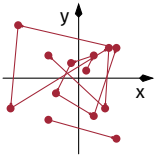


Collecting Semantics

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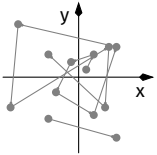


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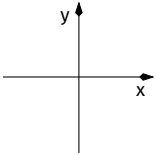


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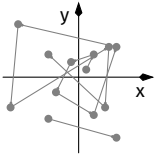


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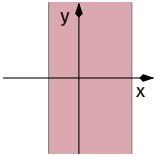


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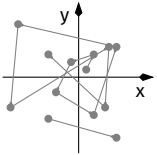


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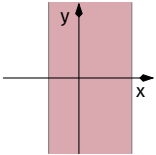


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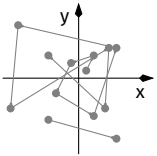


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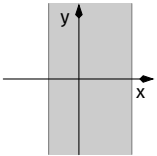


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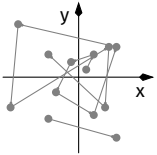


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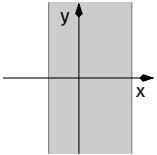


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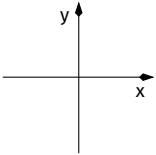


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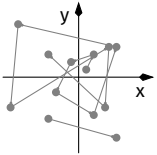


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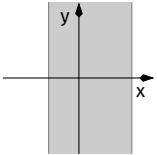


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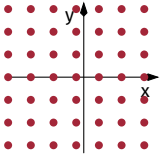


Simple Congruences

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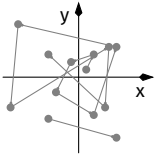


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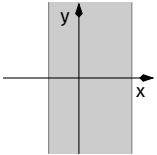


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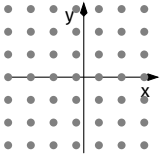


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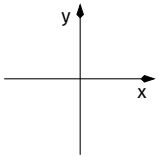


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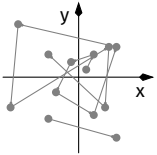


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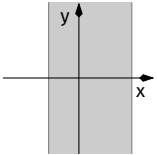


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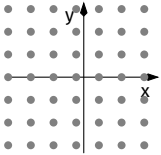


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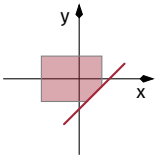


Pentagons

"Principles of Abstract Interpretation" [Cou21], "Pentagons: a weakly relational abstract domain for the efficient validation of array accesses" [LF08, p. 25]

# Abstract Domains

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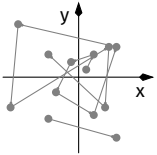


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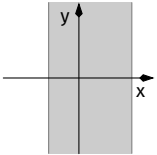


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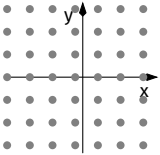


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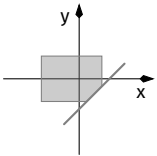


Pentagons

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# Abstract Domains

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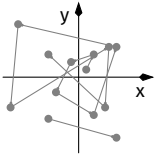


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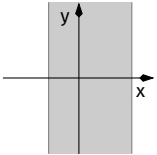


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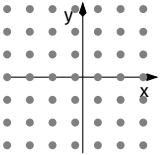


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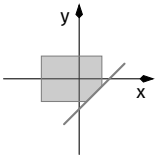



Pentagons

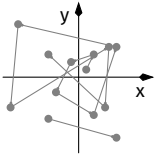
- Octagons

"Principles of Abstract Interpretation" [Cou21], "Pentagons: a weakly relational abstract domain for the efficient validation of array accesses" [LF08, p. 25]

# Abstract Domains

Pick Your Favorite Lattice!

## Numerical

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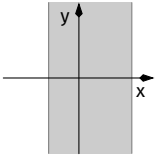


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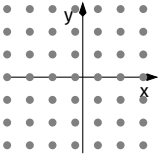


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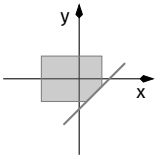


Pentagons

- Octagons
- Ellipses

"Principles of Abstract Interpretation" [Cou21], "Pentagons: a weakly relational abstract domain for the efficient validation of array accesses" [LF08, p. 25]

# Abstract Domains

Pick Your Favorite Lattice!

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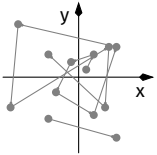
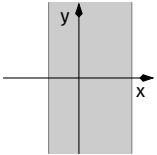



Collecting Semantics

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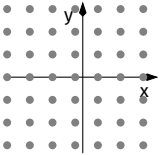


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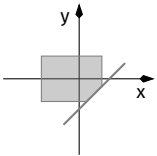



Pentagons

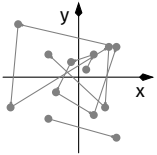
- Octagons
- Ellipses
- Exponentials

"Principles of Abstract Interpretation" [Cou21], "Pentagons: a weakly relational abstract domain for the efficient validation of array accesses" [LF08, p. 25]

# Abstract Domains

Pick Your Favorite Lattice!

## Numerical

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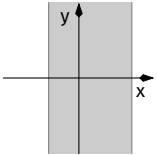


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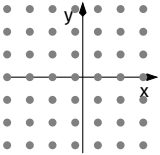


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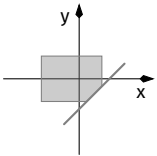


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# Sign Analysis

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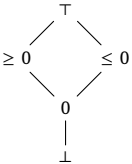

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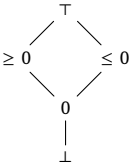
java



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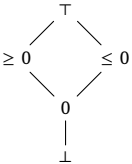

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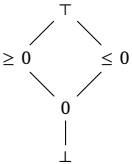

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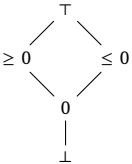

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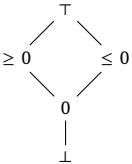

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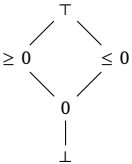

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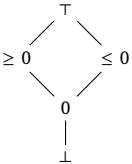

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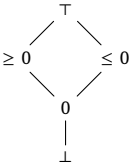

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The abstract domain we choose, e.g., intervals

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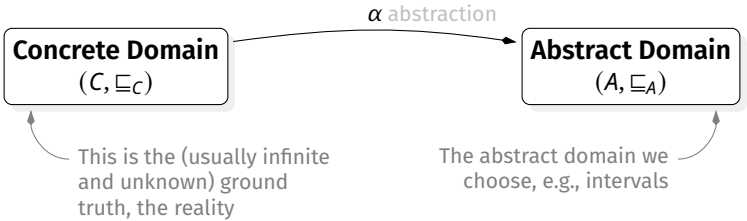

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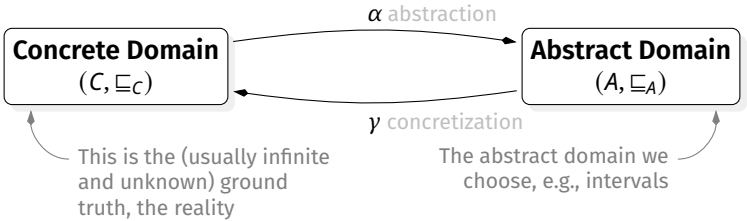

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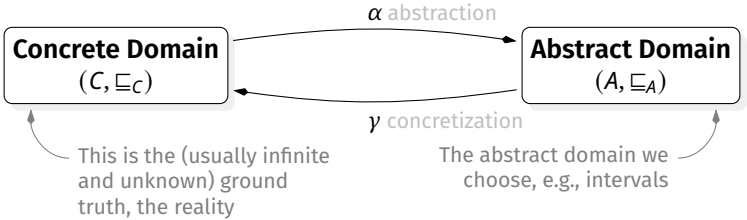


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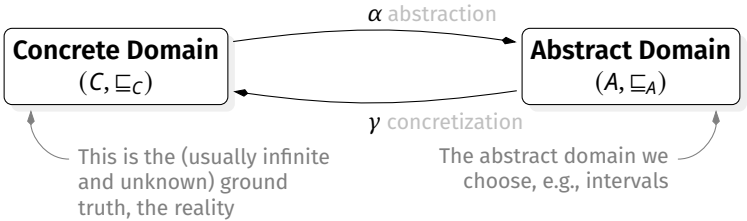


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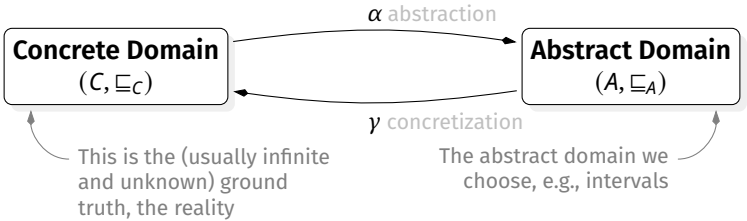


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**Évariste Galois (1811–32)**
Public Domain
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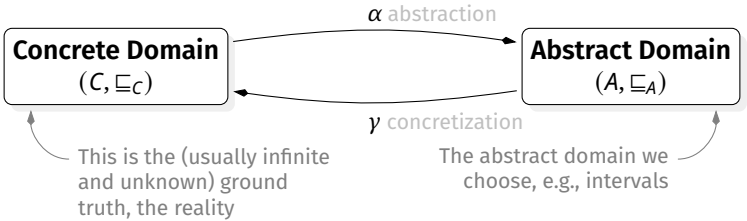


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- But, how do we handle loops? Recursion?

# Fixpoints

"A lattice-theoretical fixpoint theorem and its applications." [Tar55], "Introduction to metamathematics" [Kle52], "Principles of Abstract Interpretation" [Cou21, p. 165]

# Fixpoints

- For operators $f : X \to X$ a **fixpoint** is a $x \in X$ such that $f(x) = x$

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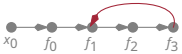
  - 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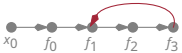
  - 

    reach a cycle, $f^{p+\ell} = f^p, \ell > 0$

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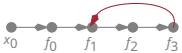
  - 

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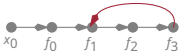
  - 

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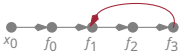
  - 

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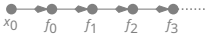
  - 

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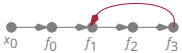
  - 

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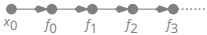
  - 

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for complete, nonempty lattices
Tarski's Theorem

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🅭🅯🄏 2.0 Oberwolfach Photo Collection

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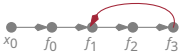
  - 

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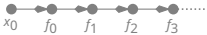
  - 

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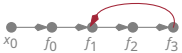
  - 

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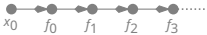
  - 

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- Analyzing, e.g. loops, we "go up" the lattice until we reach a least fixpoint

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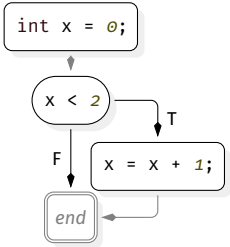

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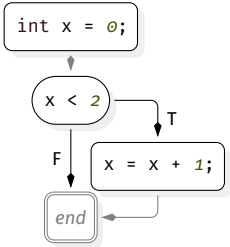


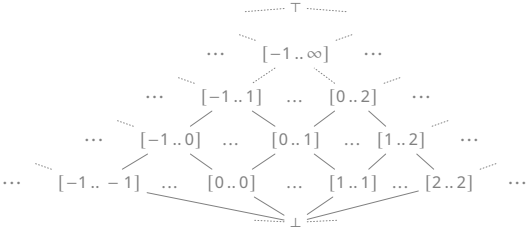

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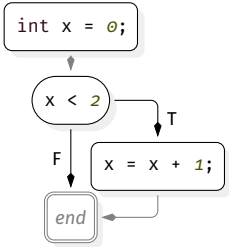


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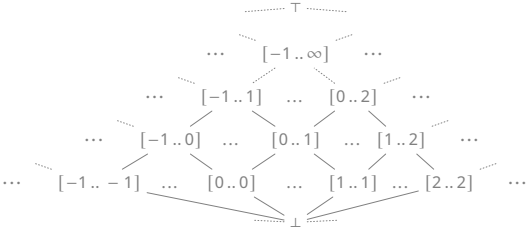

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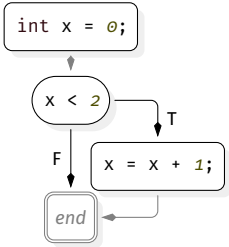


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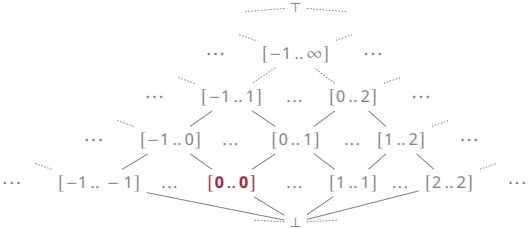

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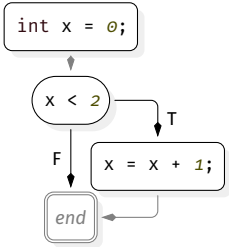


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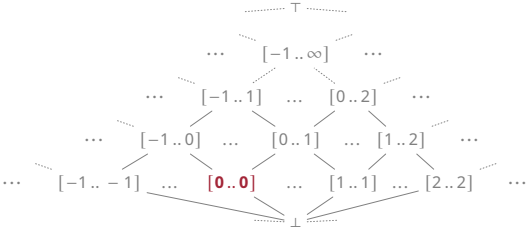

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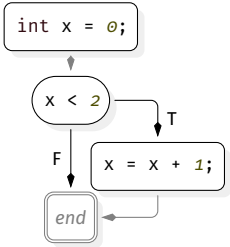


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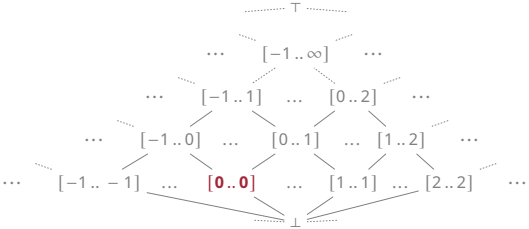

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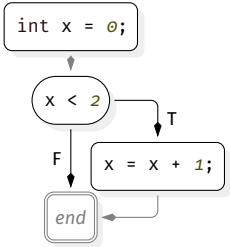


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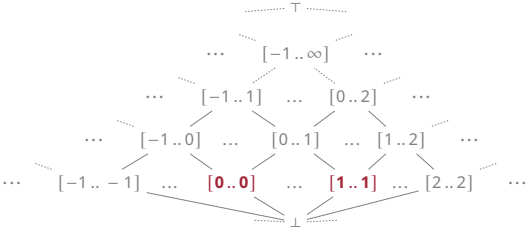

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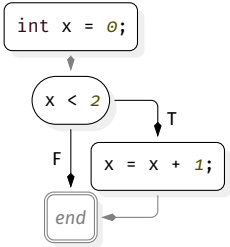


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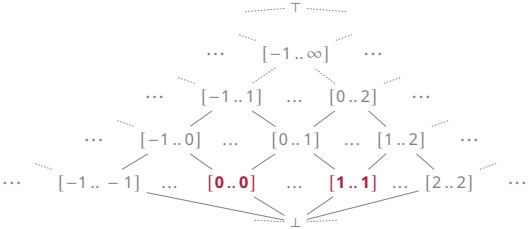

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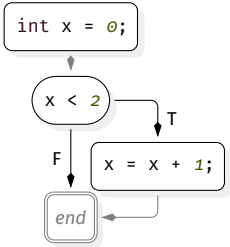


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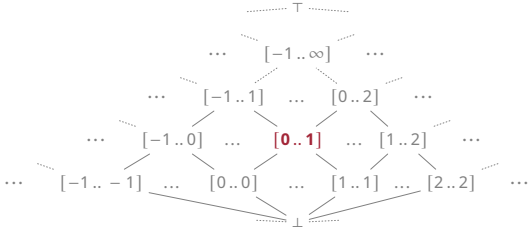

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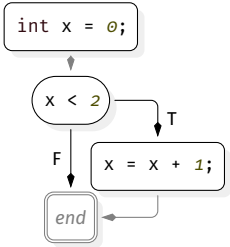


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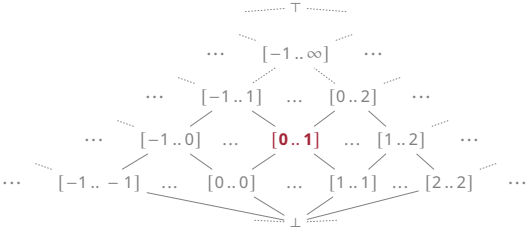

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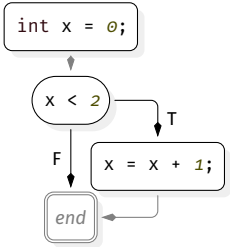


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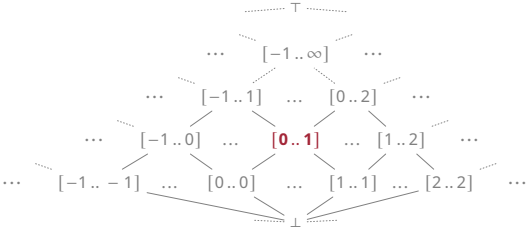

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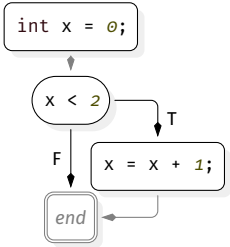


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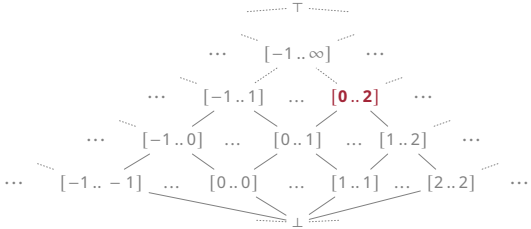

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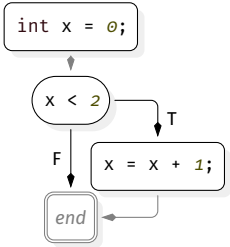


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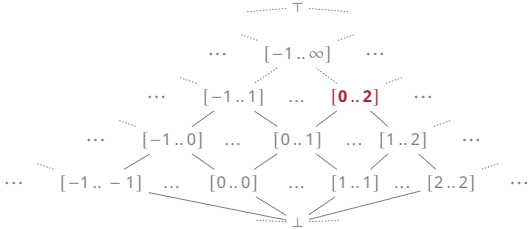

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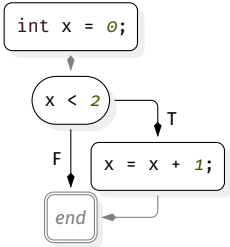
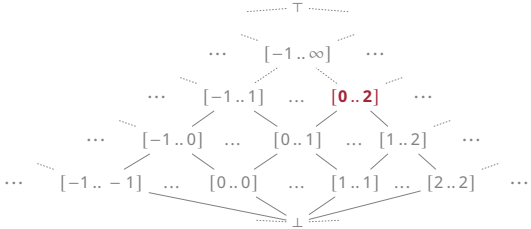
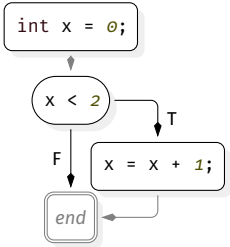



| Intervals | Signs |
|---|---|
| $\wr\, x_0 \in [0\,..\,0] \,\wr$ | $\wr\, x_0 = 0 \,\wr$ |
| $\wr$ [pre] $x_1 \in [0\,..\,2] \quad ([0\,..\,1] \cup [1\,..\,2]) \,\wr$ | $\wr$ [pre] $x_1 \geq 0 \,\wr$ |
| $\wr$ [in] $x_2 \in [0\,..\,1] \quad ([0\,..\,1] \cap (-\infty\,..\,1]) \,\wr$ | $\wr$ [in] $x_2 \geq 0 \,\wr$ |
| $\wr\, x_3 \in [1\,..\,2] \quad ([0\,..\,1] \oplus [1\,..\,1]) \,\wr$ | $\wr\, x_3 \geq 0 \,\wr$ |
| $\wr$ [post] $x_4 \in [2\,..\,2] \quad ([0\,..\,2] \cap [2\,..\,\infty)) \,\wr$ | $\wr$ [post] $x_4 \geq 0 \,\wr$ |



# Interval Analysis, I (the intuitive approach)




**Intervals**

$\wr\, x_0 \in [0\,..\,0] \,\wr$

$\wr$ [pre] $x_1 \in [0\,..\,2] \quad ([0\,..\,1] \cup [1\,..\,2]) \,\wr$

$\wr$ [in] $x_2 \in [0\,..\,1] \quad ([0\,..\,1] \cap (-\infty\,..\,1]) \,\wr$

$\wr\, x_3 \in [1\,..\,2] \quad ([0\,..\,1] \oplus [1\,..\,1]) \,\wr$

$\wr$ [post] $x_4 \in [2\,..\,2] \quad ([0\,..\,2] \cap [2\,..\,\infty)) \,\wr$

**Signs**

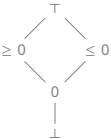
$\wr\, x_0 = 0 \,\wr$

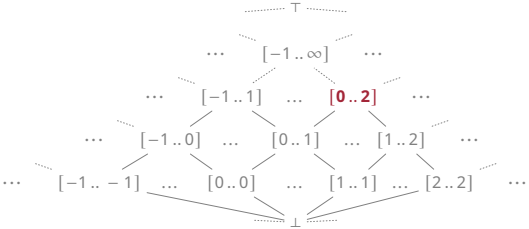
$\wr$ [pre] $x_1 \geq 0 \,\wr$

$\wr$ [in] $x_2 \geq 0 \,\wr$

$\wr\, x_3 \geq 0 \,\wr$

$\wr$ [post] $x_4 \geq 0 \,\wr$



# 3. Semantics

What does my program mean?

# Semantics

Shortened form of “Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation” [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Program Syntax (simplified)

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Program Syntax (simplified)

```java
int x = 0;

while(x < 2) {

  x = x + 1;

}
```

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Program Syntax (simplified)

Variable $v \in \mathbb{V}$

```
int x = 0;

while(x < 2) {

  x = x + 1;

}
```

java

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Program Syntax (simplified)

Variable $v \in \mathbb{V}$

Assignment

```
int x = 0;

while(x < 2) {

  x = x + 1;

}
```

java

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Program Syntax (simplified)

Variable $v \in \mathbb{V}$

Assignment

Numeric Constant $c \in \mathbb{I}$

```
int x = 0;
while(x < 2) {
  x = x + 1;
}
```

java

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Program Syntax (simplified)

Variable $v \in \mathbb{V}$

Assignment

Sequence

Numeric Constant $c \in \mathbb{I}$

```
int x = 0;

while(x < 2) {

  x = x + 1;

}
```

java

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Program Syntax (simplified)

Variable $v \in \mathbb{V}$

Assignment

Sequence

Numeric Constant $c \in \mathbb{I}$

Loop

```
int x = 0;

while(x < 2) {

  x = x + 1;

}
```

java

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Program Syntax (simplified)

Variable $v \in \mathbb{V}$

Assignment

Sequence

Numeric Constant $c \in \mathbb{I}$

Loop

Comparison $\bowtie \in \{\leq, <, \ldots\}$

```
int x = 0;

while(x < 2) {

  x = x + 1;

}
```

java

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Program Syntax (simplified)

Variable $v \in \mathbb{V}$

Assignment

Sequence

Numeric Constant $c \in \mathbb{I}$

Loop

Comparison $\bowtie \in \{\leq, <, \ldots\}$

Binary Expression

```
int x = 0;
while(x < 2) {
  x = x + 1;
}
```

java

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Program Syntax (simplified)

```
int x = 0;
while(x < 2) {
  x = x + 1;
}
```

Variable $v \in \mathbb{V}$

Assignment

Sequence

Numeric Constant $c \in \mathbb{I}$

Loop

Comparison $\bowtie \in \{\leq, <, \ldots\}$

Binary Expression

| | | | | |
|---|---|---|---|---|
| *stm* | ::= | $V \leftarrow$ *expr* | (assignment, $V \in \mathbb{V}$) | java |
| | \| | $stm_1$; $stm_2$ | (sequence) | |
| | \| | **while**(*cond*) { *stm* } | (loop) | |
| *expr* | ::= | $V$ | (variable, $V \in \mathbb{V}$) | |
| | \| | $c$ | (constant, $c \in \mathbb{I}$) | |
| | \| | $expr_1 \diamond expr_2$ | (bin. expr., $\diamond \in \{+, -, \ldots\}$) | |
| *cond* | ::= | $b$ | (boolean, $b \in \mathbb{B}$) | |
| | \| | $expr_1 \bowtie expr_2$ | (comparison, $\bowtie \in \{\leq, <, \ldots\}$) | |

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

| | | | |
|---|---|---|---|
| $expr$ | $::=$ | $V$ | (variable, $V \in \mathbb{V}$) |
| | $\mid$ | $c$ | (constant, $c \in \mathbb{I}$) |
| | $\mid$ | $expr_1 \diamond expr_2$ | (bin. expr., $\diamond \in \{+, -, \ldots\}$) |

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- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

| | | | |
|---|---|---|---|
| $expr$ | $::=$ | $V$ | (variable, $V \in \mathbb{V}$) |
| | $\mid$ | $c$ | (constant, $c \in \mathbb{I}$) |
| | $\mid$ | $expr_1 \diamond expr_2$ | (bin. expr., $\diamond \in \{+, -, \ldots\}$) |

Variable

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

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# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

| | | | |
|---|---|---|---|
| $expr$ | $::=$ | $V$ | (variable, $V \in \mathbb{V}$) |
| | $\mid$ | $c$ | (constant, $c \in \mathbb{I}$) |
| | $\mid$ | $expr_1 \diamond expr_2$ | (bin. expr., $\diamond \in \{+, -, \ldots\}$) |

Variable

Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Atomic Expression Semantics

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while(x < 2) {
   x = x + 1;
}
```

java

| | | | |
|---|---|---|---|
| $expr$ | $::=$ | $V$ | (variable, $V \in \mathbb{V}$) |
| | $\mid$ | $c$ | (constant, $c \in \mathbb{I}$) |
| | $\mid$ | $expr_1 \diamond expr_2$ | (bin. expr., $\diamond \in \{+, -, \ldots\}$) |

Variable

Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

| $\mathbb{V}$ | $\mathbb{I}$ |
|---|---|
| $x$ | 0 |
| $c$ | 5 |

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

| | | | |
|---|---|---|---|
| $expr$ | $::=$ | $V$ | (variable, $V \in \mathbb{V}$) |
| | $\mid$ | $c$ | (constant, $c \in \mathbb{I}$) |
| | $\mid$ | $expr_1 \diamond expr_2$ | (bin. expr., $\diamond \in \{+, -, \ldots\}$) |

Variable

Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

| $\mathbb{V}$ | $\mathbb{I}$ |
|---|---|
| $x$ | 0 |
| $c$ | 5 |

- Now we can define evalExpr($expr$, $env$) for an environment $env \in \mathcal{E}$

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

| | | | |
|---|---|---|---|
| $expr$ | $::=$ | $V$ | (variable, $V \in \mathbb{V}$) |
| | $\mid$ | $c$ | (constant, $c \in \mathbb{I}$) |
| | $\mid$ | $expr_1 \diamond expr_2$ | (bin. expr., $\diamond \in \{+, -, \ldots\}$) |

Variable — Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

Usually written as $\mathbb{E}[\![\, expr \,]\!]\rho$

- Now we can define evalExpr($expr$, $env$) for an environment $env \in \mathcal{E}$

| $\mathbb{V}$ | $\mathbb{I}$ |
|---|---|
| $x$ | 0 |
| $c$ | 5 |

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# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

| | | | |
|---|---|---|---|
| $expr$ | $::=$ | $V$ | (variable, $V \in \mathbb{V}$) |
| | $\mid$ | $c$ | (constant, $c \in \mathbb{I}$) |
| | $\mid$ | $expr_1 \diamond expr_2$ | (bin. expr., $\diamond \in \{+, -, \ldots\}$) |

Variable

Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

| $\mathbb{V}$ | $\mathbb{I}$ |
|---|---|
| $x$ | 0 |
| $c$ | 5 |

Usually written as $\mathbb{E}[\![\, expr \,]\!]\rho$

- Now we can define evalExpr($expr$, $env$) for an environment $env \in \mathcal{E}$

$$\text{evalExpr}(V,\ env) \stackrel{\text{def}}{=} env(V)$$

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

| | | | |
|---|---|---|---|
| $expr$ | $::=$ | $V$ | (variable, $V \in \mathbb{V}$) |
| | $\mid$ | $c$ | (constant, $c \in \mathbb{I}$) |
| | $\mid$ | $expr_1 \diamond expr_2$ | (bin. expr., $\diamond \in \{+, -, \ldots\}$) |

Variable

Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

| $\mathbb{V}$ | $\mathbb{I}$ |
|---|---|
| $x$ | 0 |
| $c$ | 5 |

Usually written as $\mathbb{E}[\![\, expr \,]\!]\rho$

- Now we can define evalExpr($expr$, $env$) for an environment $env \in \mathcal{E}$

$$\text{evalExpr}(V,\ env) \stackrel{\text{def}}{=} env(V)$$

Value of $V \in \mathbb{V}$ in Environment $env$

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

| | | | |
|---|---|---|---|
| *expr* | ::= | $V$ | (variable, $V \in \mathbb{V}$) |
| | \| | $c$ | (constant, $c \in \mathbb{I}$) |
| | \| | $expr_1 \diamond expr_2$ | (bin. expr., $\diamond \in \{+, -, \ldots\}$) |

Variable

Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

| $\mathbb{V}$ | $\mathbb{I}$ |
|---|---|
| $x$ | 0 |
| $c$ | 5 |

Usually written as $\mathbb{E}[\![\, expr \,]\!]\rho$

- Now we can define evalExpr($expr$, $env$) for an environment $env \in \mathcal{E}$

$$\text{evalExpr}(V,\ env) \stackrel{\text{def}}{=} env(V)$$

Value of $V \in \mathbb{V}$ in Environment $env$

$$\text{evalExpr}(c,\ env) \stackrel{\text{def}}{=} c$$

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

| | | | |
|---|---|---|---|
| $expr$ | $::=$ | $V$ | (variable, $V \in \mathbb{V}$) |
| | $\mid$ | $c$ | (constant, $c \in \mathbb{I}$) |
| | $\mid$ | $expr_1 \diamond expr_2$ | (bin. expr., $\diamond \in \{+, -, \ldots\}$) |

Variable — Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

| $\mathbb{V}$ | $\mathbb{I}$ |
|---|---|
| $x$ | 0 |
| $c$ | 5 |

Usually written as $\mathbb{E}[\![\, expr \,]\!]\rho$

- Now we can define evalExpr($expr$, $env$) for an environment $env \in \mathcal{E}$

$$\text{evalExpr}(V,\ env) \stackrel{\text{def}}{=} env(V)$$

Value of $V \in \mathbb{V}$ in Environment $env$

$$\text{evalExpr}(c,\ env) \stackrel{\text{def}}{=} c$$

$$\text{evalExpr}(expr_1 + expr_2,\ env) \stackrel{\text{def}}{=} \text{evalExpr}(expr_1,\ env) + \text{evalExpr}(expr_2,\ env)$$

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

$$
\begin{array}{lcll}
\mathit{expr} & ::= & V & (\text{variable}, V \in \mathbb{V}) \\
 & | & c & (\text{constant}, c \in \mathbb{I}) \\
 & | & \mathit{expr}_1 \diamond \mathit{expr}_2 & (\text{bin. expr.}, \diamond \in \{+, -, \ldots\})
\end{array}
$$

Variable

Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

| $\mathbb{V}$ | $\mathbb{I}$ |
|---|---|
| $x$ | 0 |
| $c$ | 5 |

Usually written as $\mathbb{E}[\![\, \mathit{expr} \,]\!]\rho$

- Now we can define evalExpr($\mathit{expr}$, $\mathit{env}$) for an environment $\mathit{env} \in \mathcal{E}$

$$
\begin{aligned}
\text{evalExpr}(V,\ \mathit{env}) &\stackrel{\text{def}}{=} \mathit{env}(V) \\
\text{evalExpr}(c,\ \mathit{env}) &\stackrel{\text{def}}{=} c \\
\text{evalExpr}(\mathit{expr}_1 + \mathit{expr}_2,\ \mathit{env}) &\stackrel{\text{def}}{=} \text{evalExpr}(\mathit{expr}_1,\ \mathit{env}) + \text{evalExpr}(\mathit{expr}_2,\ \mathit{env}) \\
&\vdots
\end{aligned}
$$

Value of $V \in \mathbb{V}$ in Environment $\mathit{env}$

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

$$
\begin{array}{lcll}
\mathit{expr} & ::= & V & (\text{variable}, V \in \mathbb{V}) \\
 & | & c & (\text{constant}, c \in \mathbb{I}) \\
 & | & \mathit{expr}_1 \diamond \mathit{expr}_2 & (\text{bin. expr.}, \diamond \in \{+, -, \ldots\})
\end{array}
$$

Variable

Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

| $\mathbb{V}$ | $\mathbb{I}$ |
|---|---|
| $x$ | 0 |
| $c$ | 5 |

Usually written as $\mathbb{E}[\![\, \mathit{expr} \,]\!]\rho$

- Now we can define evalExpr($\mathit{expr}$, $\mathit{env}$) for an environment $\mathit{env} \in \mathcal{E}$

$$
\begin{aligned}
\text{evalExpr}(V,\ \mathit{env}) &\stackrel{\text{def}}{=} \mathit{env}(V) \\
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\text{evalExpr}(\mathit{expr}_1 + \mathit{expr}_2,\ \mathit{env}) &\stackrel{\text{def}}{=} \text{evalExpr}(\mathit{expr}_1,\ \mathit{env}) + \text{evalExpr}(\mathit{expr}_2,\ \mathit{env}) \\
&\vdots
\end{aligned}
$$

Value of $V \in \mathbb{V}$ in Environment $\mathit{env}$

- Additionally, we can define evalCond($\mathit{cond}$, $\mathit{envs}$) and evalStm($\mathit{stm}$, $\mathit{envs}$)

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# Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
```

java

$$
\begin{array}{lcll}
\mathit{expr} & ::= & V & (\text{variable}, V \in \mathbb{V}) \\
 & | & c & (\text{constant}, c \in \mathbb{I}) \\
 & | & \mathit{expr}_1 \diamond \mathit{expr}_2 & (\text{bin. expr.}, \diamond \in \{+, -, \ldots\})
\end{array}
$$

| $\mathbb{V}$ | $\mathbb{I}$ |
|---|---|
| $x$ | 0 |
| $c$ | 5 |

Variable — Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

Usually written as $\mathbb{E}[\![\, \mathit{expr} \,]\!]\rho$

- Now we can define evalExpr($\mathit{expr}$, $\mathit{env}$) for an environment $\mathit{env} \in \mathcal{E}$

$$
\begin{aligned}
\text{evalExpr}(V,\ \mathit{env}) &\stackrel{\text{def}}{=} \mathit{env}(V) \\
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&\ \vdots
\end{aligned}
$$

Value of $V \in \mathbb{V}$ in Environment $\mathit{env}$

$\mathbb{C}[\![\, \mathit{cond} \,]\!]\mathcal{D}$

- Additionally, we can define evalCond($\mathit{cond}$, $\mathit{envs}$) and evalStm($\mathit{stm}$, $\mathit{envs}$)

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Atomic Expression Semantics

```
int x = 0;
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   x = x + 1;
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```

java

$$
\begin{array}{lcll}
\mathit{expr} & ::= & V & (\text{variable}, V \in \mathbb{V}) \\
 & | & c & (\text{constant}, c \in \mathbb{I}) \\
 & | & \mathit{expr}_1 \diamond \mathit{expr}_2 & (\text{bin. expr.}, \diamond \in \{+, -, \ldots\})
\end{array}
$$

| $\mathbb{V}$ | $\mathbb{I}$ |
|---|---|
| $x$ | 0 |
| $c$ | 5 |

Variable — Integer Values

- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

Usually written as $\mathbb{E}[\![\, \mathit{expr} \,]\!]\rho$

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$$
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\text{evalExpr}(c,\ \mathit{env}) &\stackrel{\text{def}}{=} c \\
\text{evalExpr}(\mathit{expr}_1 + \mathit{expr}_2,\ \mathit{env}) &\stackrel{\text{def}}{=} \text{evalExpr}(\mathit{expr}_1,\ \mathit{env}) + \text{evalExpr}(\mathit{expr}_2,\ \mathit{env}) \\
&\ \vdots
\end{aligned}
$$

Value of $V \in \mathbb{V}$ in Environment $\mathit{env}$

$\mathbb{C}[\![\, \mathit{cond} \,]\!]\mathcal{D}$ — $\mathbb{S}[\![\, \mathit{stm} \,]\!]\mathcal{D}$

- Additionally, we can define evalCond($\mathit{cond}$, $\mathit{envs}$) and evalStm($\mathit{stm}$, $\mathit{envs}$)

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

# Denotational Semantics while loops

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Usually written as $\mathbb{S}^\#, \mathbb{C}^\#, \mathbb{E}^\#, \ldots$

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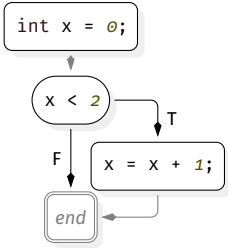


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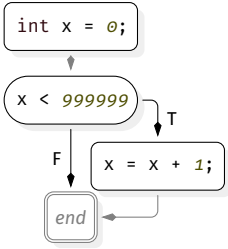


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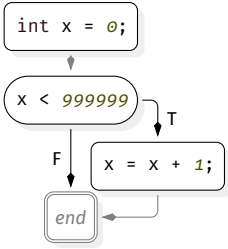


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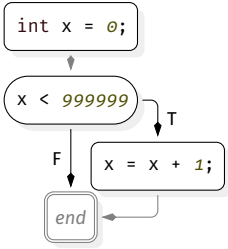


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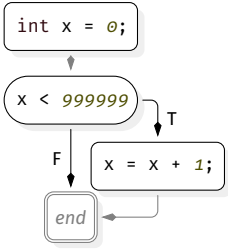


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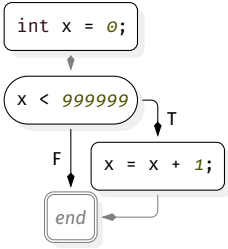


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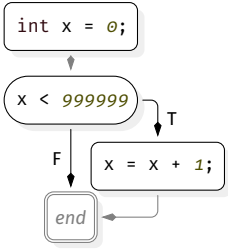


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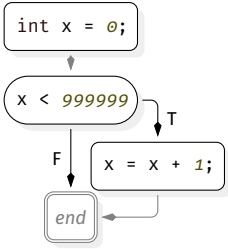


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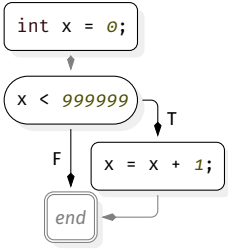


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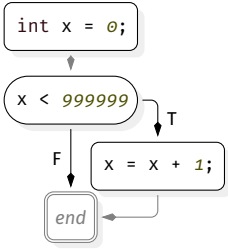


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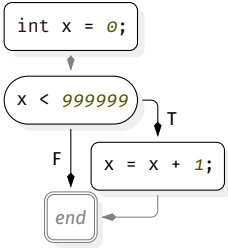


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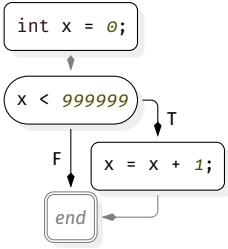


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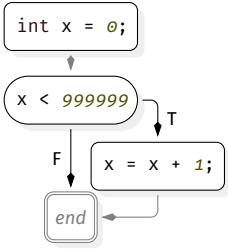


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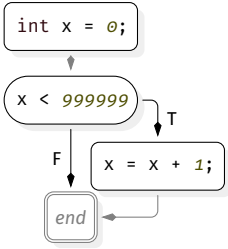


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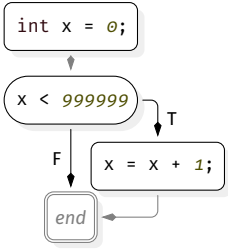


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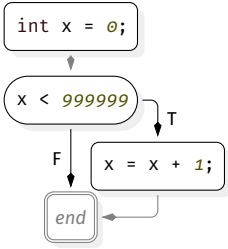


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# Let’s Bring it All Together

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\begin{array}{lcll}
\mathit{expr} & ::= & c & \text{(constant, } c \in \mathbb{R}\text{)} \\
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   | $\oplus$ | $\bot$ | $0$ | $\leq$ | $\geq$ | $\top$ |
   |---|---|---|---|---|---|
   | $\bot$ | $\bot$ | $\bot$ | $\bot$ | $\bot$ | $\bot$ |
   | $0$ | | $0$ | $\leq$ | $\geq$ | $\top$ |
   | $\leq$ | | | $\leq$ | $\top$ | $\top$ |
   | $\geq$ | | | | $\geq$ | $\top$ |
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   $x \oplus y = y \oplus x$

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|---|---|---|---|---|---|
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| $0$ | | $0$ | $\leq$ | $\geq$ | $\top$ |
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# Let's Bring it All Together — Sign Analysis

I want to make a sign analysis! (amazing!)

$$\begin{tikzcd} & \geq 0 \arrow[dr, dash] & \\ \top \arrow[ur, dash] \arrow[dr, dash] & & 0 \arrow[r, dash] & \bot \\ & \leq 0 \arrow[ur, dash] & \end{tikzcd}$$

1. Define the lattice! (there are many solutions)
   *drawing the hasse diagram is enough for us here*
   $\mathcal{L} = (x = \{\bot, 0, \leq 0, \geq 0, \top\},\ \leq = \{(\bot, 0), (\bot, \leq 0), \ldots\}, \sqcup = \{(\bot, x) \mapsto x, (0, \leq 0) \mapsto \leq 0, \ldots\},$
   $\sqcap = \{(\top, x) \mapsto x, (0, \leq 0) \mapsto 0, \ldots\}, \bot, \top)$

2. Define the abstract semantics on the following language!

   $$\begin{array}{lcll} \mathit{expr} & ::= & c & \text{(constant, } c \in \mathbb{R}\text{)} \\ & \mid & \mathit{expr}_1 + \mathit{expr}_2 & \text{(addition)} \end{array}$$

   $$\mathbb{E}[\![\, c \,]\!]\rho \stackrel{\text{def}}{=} \begin{cases} \geq 0 & \text{if } c > 0 \\ 0 & \text{if } c = 0 \\ \leq 0 & \text{if } c < 0 \end{cases}$$

   $$\mathbb{E}[\![\, a + b \,]\!]\rho \stackrel{\text{def}}{=} \mathbb{E}[\![\, a \,]\!]\rho \oplus \mathbb{E}[\![\, b \,]\!]\rho$$

   | $\oplus$ | $\bot$ | $0$ | $\leq$ | $\geq$ | $\top$ |
   |---|---|---|---|---|---|
   | $\bot$ | $\bot$ | $\bot$ | $\bot$ | $\bot$ | $\bot$ |
   | $0$ | | $0$ | $\leq$ | $\geq$ | $\top$ |
   | $\leq$ | | | $\leq$ | $\top$ | $\top$ |
   | $\geq$ | | | | $\geq$ | $\top$ |
   | $\top$ | | | | | $\top$ |

   $x \oplus y = y \oplus x$

3. Provide a concretization for $\gamma(0)$ and $\gamma(\leq 0)$!
   $\gamma(0) = \{0\}, \quad \gamma(\leq 0) = \{n \in \mathbb{Z} \mid n \leq 0\}$

4. Do we need widening? If so, define $\nabla$!
   *We do not need widening here as our lattice is finite and our semantics do not introduce new elements. (We will concretize this next week.)*

# 4. Outlook and Comments

This is incredible, I need more!

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- However, we have only looked at single variables and single domains so far!

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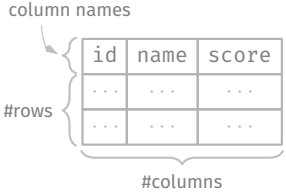

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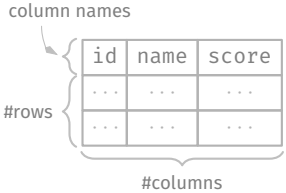

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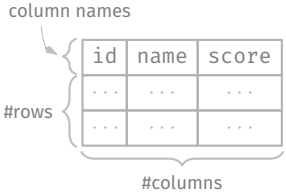

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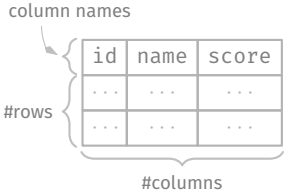

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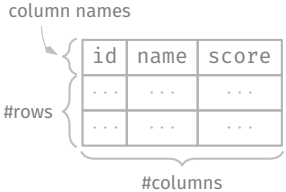

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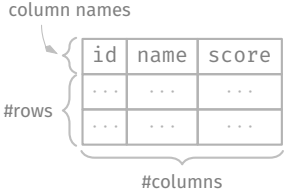


Category theory is amazing!

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column names

| id | name | score |
|---|---|---|
| ... | ... | ... |
| ... | ... | ... |

#rows

#columns

[Ger25] Oliver Gerstl. Tracking the shape of data frames in R programs using abstract interpretation (Ulm University, 2025)

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Domains can also capture relations between variables (e.g. *polyhedra*), their provenance, and much more!

However, this implies trade-offs which we discuss next time.

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2. How would you phrase that one property is “better” than another?
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